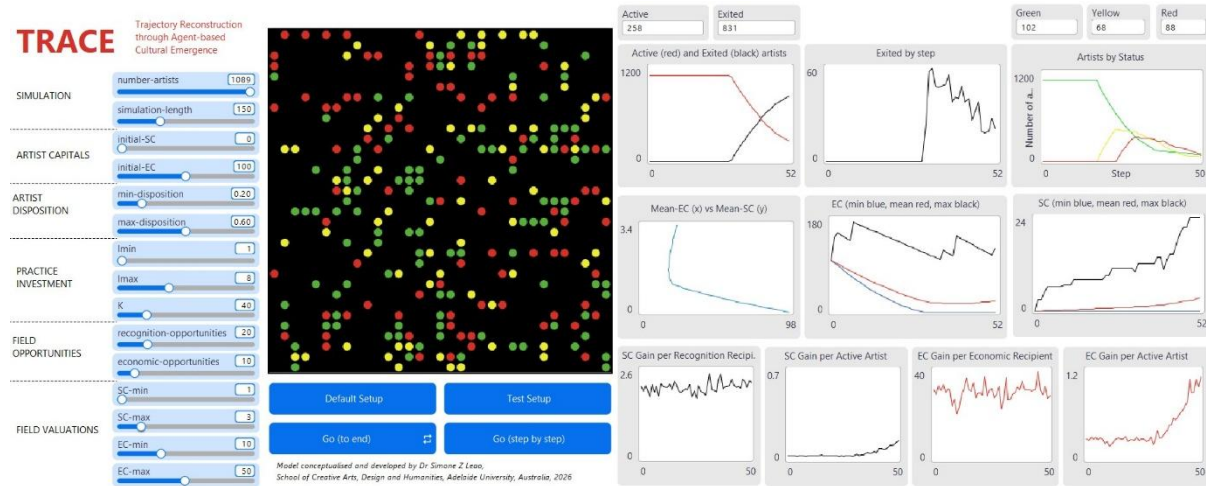


TRACE

Trajectory Reconstruction through Agent-based Cultural Emergence



Model	TRACE - Trajectory Reconstruction through Agent-based Cultural Emergence
Type	Agent-based model
Version	1.0 (Baseline Model)
Release	06 August 2026
Platform	Netlogo 7.0.4
Author	Dr Simone Zarpelon Leao
Purpose	Research and academic use
License	CC BY-NC 4.0
DOI	10.5281/zenodo.21813463

1. Overview

TRACE is an agent-based model developed to explore how artistic career trajectories emerge through interactions between symbolic capital, economic capital, individual investment behaviour, and the structure of cultural opportunities and valuation. Grounded in Bourdieu's theory of practice, the model operationalises key concepts of capital accumulation, recognition, and field dynamics to investigate the conditions shaping artistic career trajectories.

The TRACE model, code, documentation, and associated resources are publicly available through the GitHub repository via Zenodo:

<https://doi.org/10.5281/zenodo.21813463>

2. Research Questions

TRACE addresses the following research questions:

- a. How do artists accumulate symbolic and economic capital?
- b. How do unequal opportunities generate differentiated career trajectories?
- c. Under what conditions do artists persist on or exit from the cultural field?

3. Theoretical foundation

TRACE is grounded in Pierre Bourdieu's theory of practice, which conceptualises artistic careers as trajectories shaped by the interaction between individual dispositions, available resources, and the structure of the cultural field. The model translates key Bourdieusian concepts into computational elements to explore how artists accumulate, convert, and lose forms of capital over time.

Within TRACE, the artistic field is represented as a competitive environment where artists seek recognition and economic opportunities. Symbolic capital (SC) represents accumulated recognition, reputation, and legitimacy within the field, while economic capital (EC) represents the material resources required to sustain artistic practice. Artists' investment behaviour is represented through an individual disposition parameter, reflecting differences in the propensity to invest effort and resources into artistic practice. Recognition and economic opportunities represent field-level mechanisms through which capital is accumulated and converted, while withdrawal from the field represents the loss of capacity to continue artistic activity.

The model therefore operationalises Bourdieu's theory by representing artistic careers as dynamic trajectories emerging from interactions between individual dispositions, capital accumulation processes, and unequal opportunity structures.

4. Model description

TRACE follows an agent-based modelling approach in which individual artists interact within a competitive cultural field to generate emergent career trajectories. The model is structured according to the ODD (Overview, Design concepts, Details) protocol and represents the dynamic relationship between artistic investment, competition, symbolic recognition, economic opportunities, and career persistence or withdrawal.

4.1 Agents and environment

The model represents artists as autonomous agents operating within a closed artistic field. Each artist is characterised by economic capital (EC), symbolic capital (SC), an investment disposition parameter representing differences in practice-oriented behaviour, and a state variable indicating whether the artist remains active or has exited the field. The environment represents a competitive cultural field with a fixed number of recognition and economic opportunities available at each simulation step.

4.2 Processes:

At each simulation step, every active artist progresses through four sequential processes:

- a. **investment in artistic practice:** active artists first invest economic resources into artistic practice. The investment depends on their disposition to invest, minimum required investment, and maximum allowed investment. It is also influenced by an investment scaling parameter which reduces the investment if the artist economic capital is low.
- b. **allocation of recognition opportunities and symbolic gains:** the field makes a certain number of recognition opportunities available to artists at every step. They are scarce. Artists compete for those through a weighted random selection process. The weighting is based on the symbolic capital profile of the field. Artists with higher symbolic capital are more likely but are not guaranteed to be selected. After artists are selected, the opportunities are valued through a bounded randomised process. These values are accumulated to the symbolic capital of the respective artists.
- c. **allocation of economic opportunities and economic gains:** the field makes a certain number of economic opportunities available to artists at every step. They are scarce. Artists compete for those through a weighted random selection process. The weighting is based on the symbolic capital profile of the field. Artists with higher symbolic capital are more likely but are not guaranteed to be selected. After artists are selected, the opportunities are valued through a bounded randomised process. These values are added to the economic capital of the respective artists. Note that steps b and c are independent, but both allocate opportunities considering the symbolic capital of artists in the field.
- d. **evaluation of continuation in or exit from the cultural field:** artists compare their current economic capital to the required investment in their artistic practice. If the capital is enough, they remain, otherwise they exit the field. The exit is permanent and they do not compete for opportunities or gains after that.

4.3 Baseline parameters

The baseline model consists of 1,089 artists simulated over 150-time steps. All artists begin with an initial economic capital of 100 and symbolic capital of 0. Investment disposition values are randomly assigned from a range of 0.20 to 0.60, with investment amounts constrained by minimum and maximum investment values (1 to 8), and an investment scaling parameter ($K=40$). Recognition and economic opportunities are fixed at 20 and 10 per simulation step, respectively. Each recognition opportunity is valued by the field between a minimum and maximum values (1 to 3). Each economic opportunity is valued by the field between a minimum and maximum values (10 to 50). The baseline configuration represents a closed field in which artists with comparable economic and symbolic endowments compete for scarce opportunities, while career outcomes emerge from interactions between individual dispositions and field-level conditions.

5. Installation and requirements

5.1 Running TRACE in a desktop environment

TRACE requires the following software:

- NetLogo (developed and tested using NetLogo 7.0.4)
- Operating system: Windows, macOS, or Linux

No external NetLogo extensions or libraries are required to run the model.

To run TRACE, download and install NetLogo, open the TRACE_V01.nlogo model file.

TRACE_V01.nlogo model file can be found at:

https://github.com/SimoneZLeao/TRACE_ABM_model1

Analysis of simulation outputs may require additional software packages (e.g., R or Python), depending on the analytical approach adopted.

5.2 Running TRACE in a web environment

An interactive online version of TRACE is available through NetLogo Web. In this there is no need for software installation.

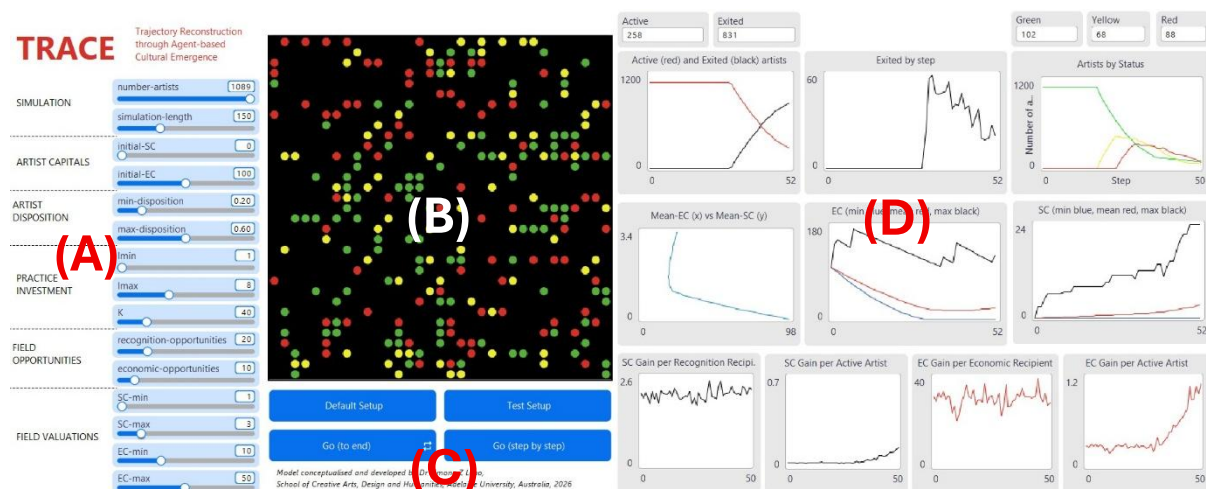
To run TRACE just click at

https://netlogoweb.org/launch#https://raw.githubusercontent.com/SimoneZLeao/TRACE_ABM_model1/refs/heads/main/TRACE_model_v1.0_NetLogoWeb.nlogox

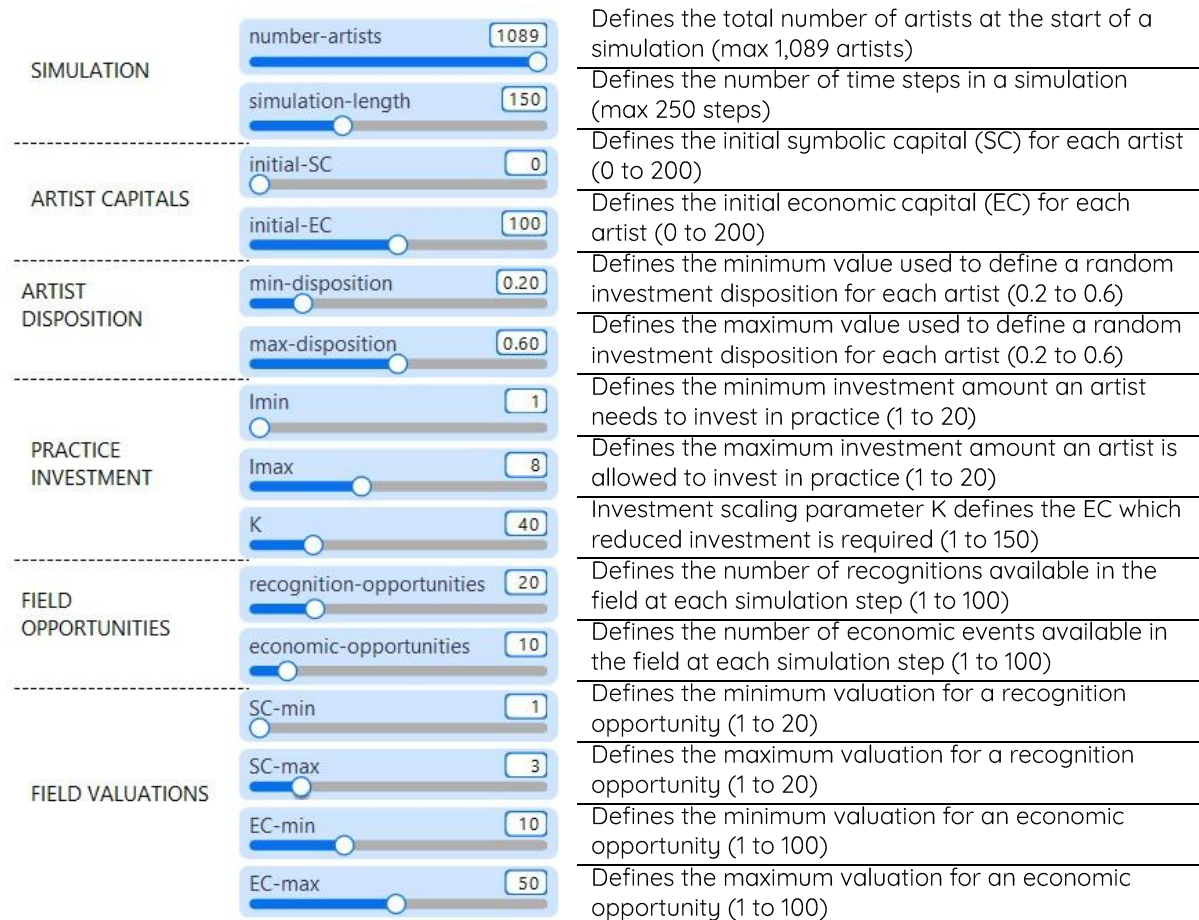
In this version, the user can run the model, visualise the simulation, and test the results if parameters are changed. However, the web environment does not allow for the outputs to be saved for future analysis.

6. Model Interface description

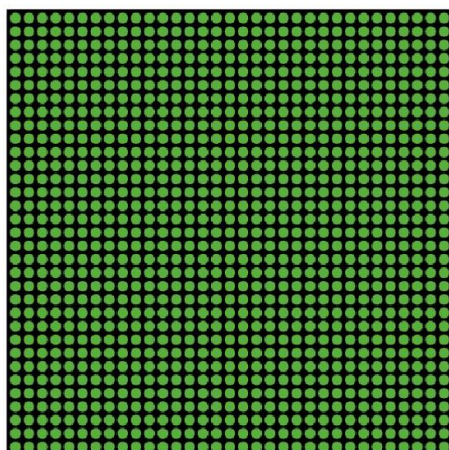
The model interface has four main parts: (a) parameters setting area; (b) simulation visualisation area; (c) processes buttons; and (d) results visualisation area. They are described in detail as follows.



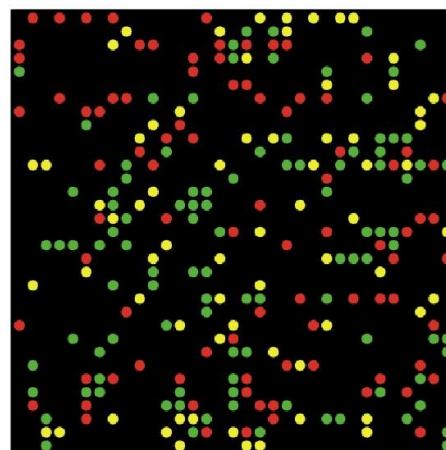
The parameters setting area (A) contains sliders for 15 model parameters. Through these sliders the user can change values for all or selected parameters within defined ranges. By changing parameters values a model user can test how the change affects the model simulation results.



The simulation visualisation area (B) is the space representing the cultural field. It is a grid with 33 rows x 33 columns. Each cell can hold only one artist (therefore the maximum 1,089 initial artists). This is an abstract space which works as an infographic, not a geographical area.



1,089 artists at simulation step 0 with EC=100



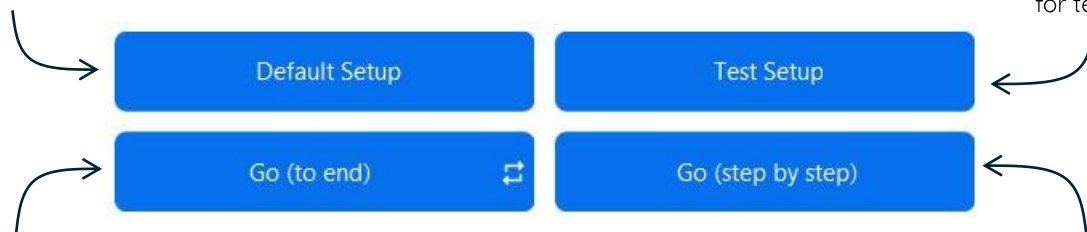
Active artists still remaining in the field at the end of the simulation, with varied financial status (secure, precarious, critical)

Artists are visualised in four colours: green (financially secure); yellow (financially precarious); red (financially critical); and black (exited from the field). The first three colours represent artists remaining active in the field.

The process buttons area (C) provides the activations for running the model. Four buttons are provided, two for setup and two for running the model, as described below.

This button sets up the model for the default values of the model parameters (baseline model)

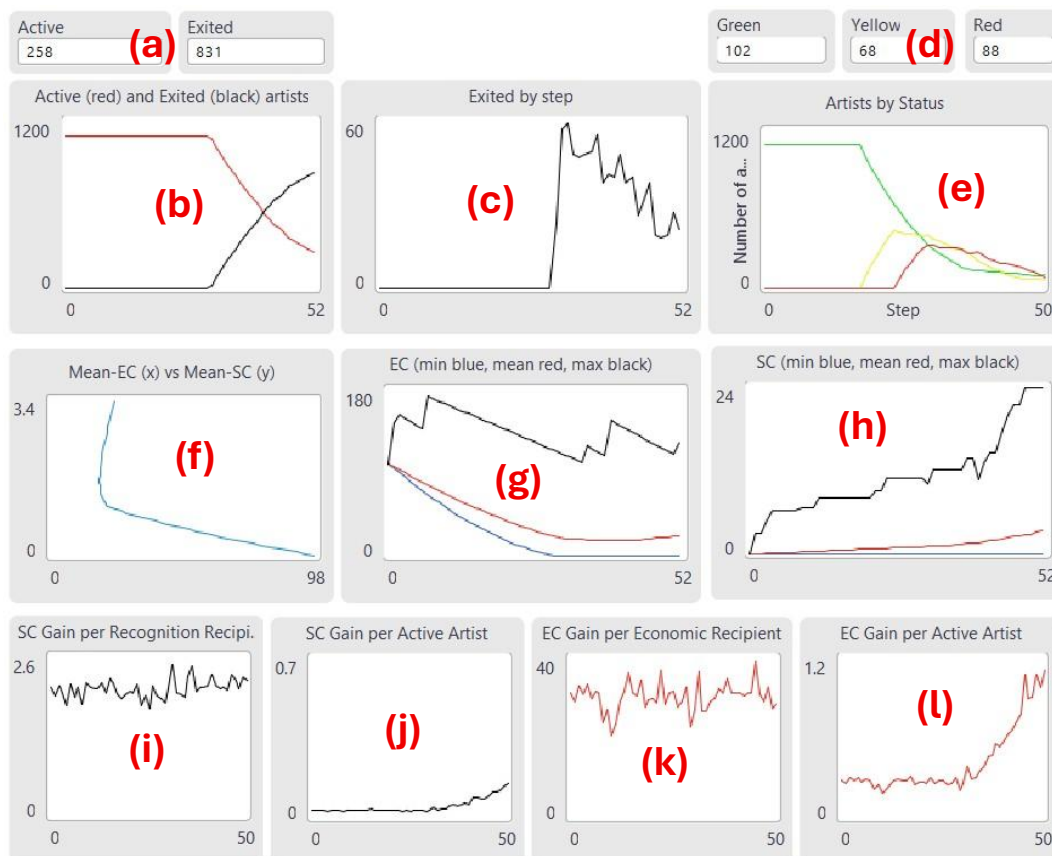
This button sets up the model for the parameter values selected by the user. This is recommended for tests.



This button runs the model from start to the last simulation step continuously. Additional clicks in this button pauses or resumes the simulation. The speed of the run can be adjusted.

This button runs the model step by step. The user must click the button multiple times for a simulation. It is useful for visualisation of results mid simulation.

The results visualisation area (D) is a dashboard with several plots and monitors which present the results of the simulation as it happens. This way the user gets an insight of how the model parameters and outputs change and vary over time.



- a. The monitors show the number of artists who are active and are exited at every simulation step. It is continuously updated to the latest step of the simulation along running.
- b. The plot shows the number of active and exited artists along the simulation, storing the progress over time.
- c. The plot shows the number of artists exited from the field at each step along the simulation.
- d. The monitors show the number of active artists who are green, yellow, or red at every simulation step. It is continuously updated to the latest step of the simulation along running.
- e. The plot shows the number of active artists along the simulation, divided in green, yellow and red status, storing the progress over time.
- f. The plot shows the relation between mean symbolic capital for all active artists at a step (y axe) and the mean economic capital for all the same artists at the same step (x axe), for each step of the simulation. It is an important plot to understand if and when a self-reinforcing process between SC and EC happens.
- g. The plot shows how economic capital minimum, mean, and maximum values for active artists vary along the simulation.
- h. The plot shows how symbolic capital minimum, mean, and maximum values for active artists vary along the simulation.
- i. The plot shows the symbolic gains by recipient of recognition opportunities at each step along the simulation.
- j. The plot shows the total symbolic gains divided by all active artists at each step along the simulation. This plot shows the decreased competition, as the number of active artists decrease as number of recognitions remain over time.
- k. The plot shows the economic gains by recipient of economic opportunities at each step along the simulation.
- l. The plot shows the total economic gains divided by all active artists at each step along the simulation. This plot shows the decreased competition, as the number of active artists decrease as number of economic opportunities remain over time.

7. Running the model

To run the baseline model just press the 'Default setup' button, then press 'Go (forever)' button or 'Go (by step)' button. Observe the results on the dashboard.

To run the model for tests, use the sliders to select parameters values, then press the 'Test Setup' button, then press 'Go (to end)' button or 'Go (step by step)' button. Observe the results on the dashboard.

8. Data output

TRACE generates individual-level simulation data in CSV format to support analysis of artist trajectories, capital accumulation, and emergent field-level patterns. Data are recorded at every simulation step for every artist, allowing researchers to reconstruct complete career trajectories over time.

The main output file, **simulation_data.csv**, contains the following variables:

Variable	Description
Tick	Simulation step (start from 0)
ArtistID	Unique identifier for each artist agent (start from 0)
EC	Artist's economic capital at the end of the simulation step
SC	Artist's symbolic capital at the end of the simulation step
State	Artist status: Active or Exited
Colour	Visual status indicator based on economic capital thresholds
InvestmentDisposition	Artist-specific investment disposition parameter
Investment	Amount of economic capital invested in artistic practice during the simulation step
RecognitionEvent	Indicates whether the artist received a recognition opportunity during the simulation step (True or False)
SCGain	Symbolic capital gained from recognition during the simulation step
EconomicEvent	Indicates whether the artist received an economic opportunity during the simulation step

The output file enables analysis of both individual trajectories and aggregate dynamics, including capital accumulation patterns, exit processes, opportunity distribution, and the emergence of differentiated career pathways. The recorded tick-by-tick data also support further analyses such as sequence analysis, clustering of career trajectories, and sensitivity analysis of model behaviour.

In addition to the individual-level simulation data, TRACE generates a configuration CSV file containing the complete set of model parameters used for each simulation run. This file provides a record of the experimental conditions under which the simulation was conducted and supports reproducibility, comparison between scenarios, and sensitivity analysis.

The configuration file, **model_configuration.csv**, contains:

Variable	Description
NumberArtists	Number of artist agents in the simulation
SimulationLength	Total number of simulation steps
RecognitionOpportunities	Number of recognition opportunities available per step
EconomicOpportunities	Number of economic opportunities available per step
InitialEC	Initial economic capital assigned to artists
InitialSC	Initial symbolic capital assigned to artists
SC-min / SC-max	Minimum and maximum symbolic capital gain from recognition events
EC-min / EC-max	Minimum and maximum economic capital gain from economic opportunities
MinDisposition / MaxDisposition	Range used to assign artist investment disposition values
Imin / Imax	Minimum and maximum investment parameters
K	Investment scaling parameter controlling the relationship between economic capital and investment amount.
T-green / T-red	Economic capital thresholds used for visual classification of artists

Together, the simulation data output and the configuration file provide a complete record of both model dynamics and parameter settings, enabling replication of experiments and systematic comparison across different model configurations.

9. Reproducibility

The results reported in the cited academic article is based on five independent simulations conducted using the baseline parameter configuration. Each simulation uses the same model settings while allowing stochastic processes within the model to generate variation in individual trajectories and aggregate outcomes. The baseline parameters and simulation outputs are provided to support replication and further analysis.

10. Model assumptions and limitations

Artists are initially homogeneous except for investment disposition. The cultural field is represented as closed, with no new artists entering the system along the simulation. Recognition allocation is simplified. Economic opportunities are externally generated. Institutional actors are not explicitly represented. Capital conversion mechanisms are simplified.

11. Future extensions:

Future developments of TRACE may extend the model to explore additional dimensions of artistic careers and cultural fields. Potential extensions include the incorporation of policy interventions, multiple interacting fields, institutional actors, alternative recognition mechanisms, demographic inequalities, and network effects. These developments would allow further investigation of how structural conditions and institutional arrangements influence the emergence of diverse career trajectories.

12. Copyright, licensing, reuse and citation

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Any use of TRACE, modified versions of TRACE, or results generated using TRACE in academic publications, reports, educational materials, or other scholarly outputs should cite the original model publications:

[To be added soon]

Commercial use of TRACE, including incorporation into commercial software, products, services, consultancy activities, or other revenue-generating applications, is not permitted under this licence. Organisations or individuals interested in commercial applications should contact the author(s) to discuss appropriate licensing arrangements.

Researchers who develop substantial extensions, adaptations, or new versions of TRACE are encouraged to notify the author(s). This supports documentation of model evolution, ensures appropriate acknowledgement of the original model, and facilitates potential collaboration within the research community.

13. Contact and collaboration

TRACE is part of an ongoing research program in computational sociology and digital humanities. The author(s) welcome the dissemination, reuse, and further development of the model and are interested in contributing to collaborative research exploring the application of computational approaches to the study of artistic careers and cultural fields.

For questions about TRACE, clarification regarding model use, or discussion of potential collaborations, please contact:

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